

slurm-temperature-check

Adversarial review of the existing package, and the rationale for reimplementing it rather than repairing it.

REVIEWED	51734bc — "Only stop slurmd and kill workload when temperature-check dies", branch onfailure-slurmd-restart
REWRITE	orphan branch rewrite, version 0.10.0
DATE	12 September 2026
FINDINGS	16 — 4 critical, 8 major, 4 minor

1 Summary

The package implements a thermal interlock: when a worker node exceeds a per-mainboard temperature threshold, its SLURM workload is killed and `slurmd` stopped. The *design intent* is sound and hard-won — a fail-closed state machine, an escape hatch on a persistent filesystem, a test hook on an ephemeral one, per-board thresholds, and an eleven-row truth table that is a genuine specification.

The *implementation* has four defects that cause the interlock to fail **open** — that is, to leave a node running jobs with no thermal protection, or to kill the wrong processes — and each of them is silent. Two were reproduced directly during this review; the other two are evident from the shipped scriptlets and the `%files` list.

Separately, the mechanism's documented enforcement path no longer exists. The README's central code block prescribes `Bindsto=temperature-check.service` on `slurmd.service`, but commit 51734bc removed that directive. The shipped drop-in is still named `10-bind.conf` and now carries only `After=`, which has no effect when the guard dies. Enforcement moved to `OnFailure=` plus a PID-sweeping oneshot unit — the same unit that harbours the most serious finding below.

2 Why a rewrite rather than repairs

Each individual defect is fixable in isolation. The reason to rewrite is that most of them are not independent: they are consequences of two structural decisions, and repairing the symptoms one at a time leaves the structure that produced them in place.

A reader process and a checker process communicating through a file

`temperature-provider` writes `/run/temperature/temperature`; `temperature-check` reads it. That file is the sole cause of the partial-read tolerance (`temperature_file_maximum_times_invalid_integer`), the dependency on the deprecated `python3-atomicwrites`, the practice of writing English error strings where a number is expected, and the staleness check that exists only to notice that the writer died. Four mechanisms, all serving one file. Once the reading is a `sysfs` read rather than a subprocess, the split buys nothing, and the "did the reader stop" question is answered properly by `systemd's WatchdogSec=`.

Shelling out to sensors and parsing its output

This requires `lm_sensors` on every node, a subprocess and a timeout per reading, a board-to-chip dispatch table written as a shell `case` statement, and a regular expression over human-readable output. It also forces the board knowledge to live in two files in two languages — `temperature-provider.sh` holds board → sensor, `temperature-`

`check.conf` holds `board` → `threshold` — kept in step by hand. `sensors` reads the same `sysfs` attributes the program can read directly, so all of that is removable at once.

Repairing within the existing structure would mean keeping two Python daemons, two board tables, four divergent spec files, a Vagrant-only build, and a `%post` scriptlet whose correctness the interlock depends on. The rewrite deletes the structure and with it eleven of the sixteen findings; the remaining five are fixed on their merits.

The secondary reason is verifiability. The truth table in the README was a specification with no executable counterpart — the `bat.s` suite that nominally covered it has a test that passes for the wrong reason (F7). In the rewrite the table *is* the test, row for row, so the specification cannot drift from the code without failing CI.

3 Findings

ID	AREA	SEVERITY	FINDING
F1	Emergency stop	CRITICAL	<code>get_slurm_pids</code> can emit truncated PIDs, which are then sent <code>SIGKILL</code>
F2	Packaging scriptlet	CRITICAL	The <code>%post</code> drop-in shuffle leaves a node unprotected if the transaction is interrupted
F3	Configuration	CRITICAL	A missing optional-looking config key raises an uncaught <code>KeyError</code> , which kills the node's jobs
F4	Packaging	CRITICAL	A fresh install ships no usable configuration, so the guard cannot start
F5	Documentation	MAJOR	The README prescribes <code>BindsTo=</code> , which the shipped drop-in no longer contains
F6	Emergency stop	MAJOR	<code>slurmd</code> is <code>SIGKILL</code> ed before <code>systemd</code> is asked to stop it
F7	Guard logic / tests	MAJOR	The staleness check cannot fire for the override file; the test covering it passes for the wrong reason
F8	Provider	MAJOR	Error prose is written into the file that is supposed to hold a number
F9	Provider	MAJOR	The temperature-matching regular expression does not express what it is meant to
F10	Packaging	MAJOR	<code>pstree</code> and <code>pgrep</code> are used but their packages are not required
F11	Structure	MAJOR	Board knowledge is duplicated across two files in two languages
F12	Packaging	MAJOR	Four spec files have drifted to three versions and none is rebuildable from an SRPM
F13	Dependencies	MINOR	<code>python3-atomicwrites</code> is deprecated upstream and replaceable with three lines of <code>stdlib</code>
F14	<code>systemd</code>	MINOR	<code>PermissionsStartOnly=</code> is a deprecated directive
F15	Repository	MINOR	5.3 MB of internal meeting slides make up most of the repository
F16	Logging	MINOR	Log files plus <code>logrotate</code> plus <code>tmpfiles</code> duplicate what the journal provides

4 Critical findings

F1 Truncated PIDs are sent SIGKILL CRITICAL

`get_slurm_pids` collects the job's processes by parsing the ASCII art that `pstree` draws:

```
L2=$(pstree -p ${i} | egrep -o '([0-9]*)')
```

`pstree` truncates long lines by default; `-l` is the opt-in to stop it doing so. When the output is truncated mid-PID, the regular expression happily harvests the surviving digits, and `emergency-stop-slurmd.service` then sends `kill -9` to that number — a real, unrelated process. Under `systemd` there is no terminal, so the fallback width applies, and a job tree wider than it is entirely ordinary for a multi-rank MPI step.

REPRODUCED

```
$ COLUMNS=40 pstree -p 1 | head -2
?(1)---claude(30)---zsh(2774)---head(27+
      |                               ^-pstree(+

$ COLUMNS=40 pstree -pl 1 | head -2      # -l: do not truncate
?(1)---claude(30)---zsh(2774)---head(2782)
      |                               ^-pstree(2781)
```

`head`'s PID is 2782. The truncated line yields 27, and `pstree --help` documents `-l`, `--long` don't truncate long lines.

Collecting a PID list and killing it a moment later is racy besides: a process that exits in between frees its PID for reuse. `Cgroup` membership answers the same question exactly, and needs no external command.

F2 The upgrade scriptlet can disarm the interlock CRITICAL

To keep a package upgrade from killing the jobs on the node, `%post` moves both drop-ins out of the unit directory, restarts the services, and moves them back:

```
rm -f /tmp/temperature-check-hack
mv /usr/lib/systemd/system/slurmd.service.d/10-bind.conf /tmp/temperature-check-hack 2>/dev/null
...
systemctl restart temperature-provider
systemctl restart temperature-check
mv /tmp/temperature-check-hack /usr/lib/systemd/system/slurmd.service.d/10-bind.conf 2>/dev/null
```

Three problems compound. If the transaction is interrupted between the two moves, the node keeps running jobs with the interlock's `OnFailure=` hook absent — unprotected, and with nothing reporting it. `/tmp/temperature-check-hack` is a predictable path in a world-writable directory, so a local user can pre-create it. And every `mv` discards its errors, so a failure at any step is invisible. The scriptlet is load-bearing for safety, which is the opposite of what a scriptlet should be.

F3 A missing configuration key kills the workload CRITICAL

`temperature_file_maximum_age_seconds` and `board_filename` are both read by `temperature-check` but absent from its `required_parameters` list. They are, however, present in `integer_parameters`, which is in-

dexed without a guard. A configuration that omits either produces an uncaught exception, a non-zero exit, and therefore `OnFailure=` — killing the jobs on the node.

REPRODUCED

```
$ python3 temperature-check -c tc-noage.conf
Traceback (most recent call last):
  File "temperature-check", line 74, in <module>
    if not isinstance(config[x], int):
        ~~~~~^
KeyError: 'temperature_file_maximum_age_seconds'
```

F4 A fresh install has no configuration to read **CRITICAL**

The `%files` list ships `/etc/temperature-check.conf.default` and `/etc/temperature-provider.conf.default`, but the programs default to `/etc/temperature-check.conf` and `/etc/temperature-provider.conf`. Nothing in the package copies one to the other; the `bats` suite does it by hand. A newly installed node therefore has a guard that exits non-zero on startup — which fires the emergency stop. In practice this is masked only by F2's scriptlet, which is precisely why that scriptlet exists. Two defects hold each other up.

5 Correctness findings

F5 The documented enforcement path was removed **MAJOR**

The README presents the mechanism as a systemd binding, with a truth table of observed behaviour to match:

```
[Unit]
BindsTo=temperature-check.service
After=temperature-check.service
```

Commit 51734bc removed `BindsTo=` and the `ExecStop=` beside it. The shipped `slurmd.service.d/10-bind.conf` now contains `After=` alone, which orders startup and does nothing when the guard dies. The file's name still says "bind". The intent behind the change — letting an operator restart `slurmd` without killing jobs — is right, and the rewrite keeps it; the documentation was simply left behind.

F6 The emergency stop is ordered backwards **MAJOR**

```
ExecStart=/usr/bin/bash -c "/usr/bin/kill -9 $(/usr/bin/get_slurm_pids); systemctl stop slurmd.service"
```

`slurmd` is `SIGKILL`ed first and systemd asked to stop it afterwards, so its shutdown never runs and `slurmctld` sees the node vanish rather than deregister. Worse, a site that has added `Restart=` to `slurmd.service` gets a restart scheduled the moment the main process is killed, racing the `stop` that follows — `slurmd` can come back up on a node that has just been declared too hot.

F7 A documented case cannot occur, and its test passes anyway **MAJOR**

Row 4 of the README's truth table specifies that an override file which is *too old* must stop the node. In the `stat_file` state, the override path takes `temperature_mtime = time.time()` — the current time — so the override file is never stale and row 4 is unimplemented.

The corresponding `bats` case, *"test 4: does temperature-check stop when the temperature_override_filename is too old?"*, nonetheless passes: the file it inherits from test 3 is empty, so the run stops on the invalid-integer path instead. A test that passes for the wrong reason is worse than a missing one, because it reports coverage that does not exist.

F8 Error prose where a number is expected MAJOR

On a timeout or a failed subprocess, `temperature-provider` writes sentences into the temperature file:

```
temperature_reading = "Error: read_temperature_command timed out."
```

The checker cannot distinguish this from a corrupt reading, so it consumes the invalid-integer budget and, at the default of two tolerated failures, stops the node after three consecutive `sensors` hiccups. Stopping is the safe direction, but the channel carries no way to tell a transient from a real fault, so the operator cannot either.

F9 The temperature regex does not say what it means MAJOR

```
find_temperature_regex: '(\d+)\.*\d+'
```

`\.*` is *zero or more dots*, not an escaped decimal point, so the pattern matches `450` as readily as `45.0`, and requires a second digit run that a whole-number reading does not have. It survives only because `temperature-provider.sh` narrows the input to a single line with `grep -A2 <chip> | tail -n1` first — two fragile steps propping each other up. The integer part is also truncated rather than rounded, so a `85.9 °C` reading is compared as `85`.

F10 Undeclared runtime dependencies MAJOR

`get_slurm_pids` calls `pstree` (`psmisc`) and `pgrep` (`procps-ng`). Neither package appears in `Requires:.` On a node where `pstree` is absent the descendant sweep silently collects nothing, and the interlock kills less than it reports.

6 Structural findings

F11 Board knowledge lives in two places MAJOR

Eight mainboards are enumerated twice: `temperature-provider.sh` maps `board` → `sensor chip` in a shell case, and `temperature-check.conf` maps `board` → `threshold` in YAML. Adding a model means editing two files in two languages, and a mismatch between them is not detectable by either program.

F12 Four spec files, three versions, no reproducible build MAJOR

`packaging/` carries `generic` (0.7.1), `centos7`, `rockylinux8` (0.9.1) and `rockylinux9` (0.9.3) spec files. The two current ones copy their inputs out of `/vagrant`, and have no `Source0` and no `%prep`, so they build only inside the Vagrant VM and cannot be rebuilt from an SRPM. CentOS 7 has been end-of-life since June 2024.

F13 A deprecated dependency MINOR

`python3-atomicwrites` is deprecated by its author. What the provider needs from it — write to a temporary file in the same directory, then rename — is `os.replace()`, three lines of standard library.

F14 A deprecated systemd directive MINOR

`temperature-provider.service` uses `PermissionsStartOnly=true`, superseded years ago by the `+` prefix on the `ExecStartPre=` line itself.

F15 The repository is mostly meeting slides MINOR

`doc/` holds 5.3 MB of the repository's 11 MB: LaTeX sources, rendered PDFs and PNG diagrams from two internal cluster meetings in 2021 and 2022. The useful content is the state diagram and the dependency structure, both of which are a paragraph and a diagram in prose.

F16 Logging reimplements the journal MINOR

Both units are `Type=simple`, so their output is journalled for free. The package nonetheless ships a log directory, a `tmpfiles.d` entry, a `logrotate.d` entry and a `--log` flag using `WatchedFileHandler`, to arrive at what the journal already does — with no `compress`, `notifempty` or `su` in the rotate configuration.

7 What was kept

The review found the package's thinking better than its code, and the rewrite is a reimplementation of that thinking rather than a fresh design:

- **Fail-closed as the governing rule.** Every path that does not end in a reading at or below the threshold ends with the node out of service.
- **The disable file** on a persistent filesystem, stated on every pass, so the interlock can be suspended without touching the unit, its enabled state or the configuration.
- **The override file** on an ephemeral filesystem, read in place of the hardware, so the whole path can be proven on a live node.
- **Per-board thresholds**, carried over with their values unchanged.
- **The TESTBOARD hook**, which keeps a verification path inside the shipped package.
- **The truth table**, promoted from prose to the test suite.
- **The decoupling intent of 51734bc** — an operator must be able to restart `slurmd` without killing jobs — now expressed as `Wants=/ After=` with enforcement on `OnFailure=` only.
- **The "situations to consider" reasoning** from the README, which is where the escape hatch, the test hook and the staleness concern came from in the first place.

8 Simplifications adopted

CHANGE	EFFECT
Read <code>/sys/class/hwmon</code> directly	Removes the <code>lm_sensors</code> dependency, the per-reading subprocess and its timeout, the shell dispatch script and the regular expression (F8, F9)
One process instead of two	Removes the intermediate file and, with it, the partial-read tolerance, the atomic-write dependency, the error-string channel and the staleness check (F7, F13)
<code>WatchdogSec=</code> with the keep-alive sent from the check loop	Answers "has the loop stopped turning" properly, which is the only thing the two-process split protected against

CHANGE	EFFECT
One board table: board → chip, sensor, limit	Removes the duplicated board knowledge (F11)
Kill by cgroup with <code>systemctl kill --kill-whom=all</code>	Removes <code>get_slurm_pids</code> , <code>pstree</code> and <code>pgrep</code> entirely (F1, F10)
One spec with a real <code>Source0</code> , built in containers	Rebuildable from its SRPM; replaces four drifted specs and the Vagrant harness (F12)
No restart on package upgrade	Removes the scriptlet the interlock's safety depended on (F2)
A shipped <code>%config(noreplace)</code> board table	A fresh install on a known board works without manual steps (F4)
Structured logging to the journal	Removes the log directory, <code>tmpfiles.d</code> and <code>logrotate.d</code> (F16)
Strict configuration parsing	Unknown keys, repeated sections, missing thresholds and implausible limits are errors, not defaults (F3)

9 Decisions taken

Implementation: Go, one static binary

The package's runtime dependency set goes to `systemd` alone — no interpreter, no YAML library, no `lm_sensors`. An interlock present on every worker node should not be able to break because an unrelated package was upgraded. The configuration parser is hand-written against the standard library, so `go.mod` carries no `require` block and CI fails if it grows one.

Enforcement: cgroup membership, not a PID list

`systemctl kill --signal=SIGKILL --kill-whom=all` on `slurmstepd.scope` and on `slurmd.service`, then `systemctl stop slurmd.service`. This is exact and race-free regardless of `KillMode=process`, covers both the cgroup/v2 scope layout of SLURM 22.05+ and the older one, and ends with the unit stopped so a `Restart=` setting cannot undo it (F1, F6, F10).

Privilege split

The guard reads world-readable `sysfs` as an unprivileged system user with an empty `CapabilityBoundingSet=`, and reports its verdict by exiting. Everything that can kill a process lives in a separate unit whose commands are fixed in the unit file and take no input from the configuration or the sensors, reachable along exactly one edge — the guard's `OnFailure=`.

Licence: Apache-2.0

Replacing GPL-3.0, with a `NOTICE` naming GSI as copyright holder and SPDX identifiers on the packaging files. All prior authors hold GSI addresses, so the copyright holder is unchanged by the relicensing.

10 Verification status

What was checked while producing the rewrite:

- `go test -race ./...` green across all six packages; `gofmt` and `go vet` clean.
- The composed chip names are pinned to the names the deployed board table already uses — `k10temp-pci-00c3` from PCI function `0000:00:18.3`, `coretemp-isa-0000`, `spd5118-i2c-20-50` — which is the closest available substitute for a capture from the real hardware.

- The binary was driven through every documented path against a fabricated `sysfs` tree: sensor below and above the limit, the limit boundary to the thousandth, the disable file, the override file, and an unknown board. Exit codes came out 0, 1 and 2 as specified.
- The SRPM builds and is self-contained; `rpmspec` expansion verified.

What remains unverified, and by what:

- **The RPM's build sections.** Only `go-srpm-macros` is installed on the review machine, not `go-rpm-macros`, so `%build`, `%install` and `%check` did not run. CI's Rocky 9 container covers them.
- **Unit verification and the hardening gate.** `systemd-analyze` and `shellcheck` were unavailable, so `systemd-analyze verify`, `systemd-analyze security --threshold=30` and the shell lint are untested locally. The first CI run is the real check.
- **Three board entries are inferences.** `H12DSG-0-CPU` and `BC15MASA` carry `chip = i350bb` with no explicit sensor, reproducing the previous `grep -A2 i350bb | tail -n1`. Confirm with `--list` on one node of each model.
- **Real heat.** No test suite can warm a mainboard. The drain-and-override procedure in the README is the acceptance test, and it has to be run on a real node.

One behavioural change deserves separate notice before rollout: thresholds are now compared against the exact reading in the millidegrees the kernel reports, rather than against a truncated integer. A node at 85.4 °C against an 85 °C limit now stops where the previous implementation did not, so any node habitually sitting at its limit will begin to trip.